1. Patient Registration (Online / First Login)

Required Identification Data:

Field	Details / Notes
Full Name	_____
Age	_____
Sex	_____
Religion	_____
Occupation	_____
Marital Status	_____
Educational Status	_____
Date of Admission	Auto-filled or chosen
Address	Country / Region / Zone / Woreda / Kebele
Date of Birth (Optional)	_____

Optional / Smart Fields:

- Pre-existing conditions
- Allergy information
- Trauma or emergency flag

Patients submit online, and a **unique Patient ID** is generated. Submission time is recorded for queuing.

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2. Queue Management & Prioritization

Smart Queuing Features:

- Patients are queued based on submission time by default.
- **Emergency prioritization:**

    - Trauma, labor, or life-threatening conditions **jump the queue** automatically.
- **Notification system:**
    - Patient receives SMS or app notification:

      "Hello [Patient Name], you are number 10 in the queue. Please be ready to be evaluated at [Hospital Name]."

- Doctors see **real-time queue updates**, including emergency tags.

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3. Doctor Portal

Dashboard View:

- Patient Identification Data (all submitted info)
- Queue position and arrival time
- Editable sections for:
    - Physical examination findings
    - Vital signs
    - Investigations / lab orders
    - Diagnosis and treatment plan

Lab Orders & Follow-up:

- Doctor can submit lab requests through the system.
- System automatically notifies the patient when results are ready:

  "Hello [Patient Name], your lab result is ready. Please contact [Hospital Name] for follow-up. Our AI assistant can summarize the findings if needed."

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4. Notification System

Types of notifications:

1. **Queue Updates:** SMS / app notification when approaching the doctor.
2. **Lab Results:** SMS / app message when results are ready.
3. **Reminders:** Follow-up, medication, or appointment reminders.
4. **Emergency Alerts:** Immediate notification if conditions change or urgent action is required.

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5. History Tracking

- All patient visits are **stored securely**.
- Includes:
  - Vitals
  - Lab results
  - Imaging and radiology reports
  - Prescriptions and procedures
- Doctors can **view previous hospitalizations** and monitor trends.

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6. Access Control

- **Doctor access is partitioned** per hospital to maintain confidentiality.
- Referrals allow selective sharing between hospitals.
- Lab technicians and administrative staff have **role-based access** to relevant data only.

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7. Smart Features / Future Add-ons

- **AI-assisted insights:** System can flag unusual results, suggest preliminary interpretations, and guide doctors.
- **Analytics dashboard:** Hospital administrators can monitor patient flow, lab workload, and wait times.
- **Mobile app for patients:** Track queue, lab results, and receive notifications without visiting the hospital.
- **Integration with national health records:** For future nationwide access.

(*Doctor fills after patient logs in; optional email/phone for notifications*)

---- B. General Appearance

---- C. Vital Signs – Mandatory

- Blood Pressure (BP)
- Pulse Rate (PR)
- Respiratory Rate (RR)
- Temperature (T°)
- SpO_2
- Body Mass Index (BMI)

- Pain Level

---- D. HEENT (Head, Eyes, Ears, Nose, Throat)

---- E. Lymphoglandular System

---- F. Respiratory System

---- G. Cardiovascular System

---- H. Abdomen (Gastrointestinal)

---- I. Genitourinary (GUT)

---- J. Musculoskeletal System (MSK)

---- K. Integumentary System

---- L. Neurological System